Nonconsecutive Subset Sums Modulo Three: Exact Encodings and an Unproved Recurrence Candidate

SIGMA AI-assisted research synthesis

Abstract

We study subsets of an initial interval of positive integers that contain no consecutive elements and whose sum is divisible by three, including the empty subset. The available results give exact counting encodings and bounded computational evidence, but not a complete proof of the requested recurrence or asymptotic density. Exact enumeration refutes the short rule $a_{n+3} = a_{n+2} + a_n$, while finite exact calculations report $a_{12} = 128$ and support an order-twelve recurrence involving every third term. A gap-removal bijection explains a second encoding by Gaussian polynomials; its weight requires careful separation of the positional shift from the standard subset-sum normalization.

The root-of-unity representation identifies the remaining mathematical tasks: justify the ordered periodic matrix products, handle the initial boundary, and bound the nontrivial modes relative to Fibonacci growth. These results narrow the problem to specific algebraic and asymptotic obligations without treating finite agreement as proof.

Evidence coverage limitation

The evidence index omitted 996 material saved research field fragment(s) across 6 source(s). These fragments were not reviewed for this paper. Interpretations that could depend on omitted material remain provisional; the complete frozen source snapshot and omission ledger are retained with the generation record. *Claims discussed:* C10, C11

1 Introduction

For each integer $n \geq 0$, let a_n count subsets $S \subseteq \{1, \ldots, n\}$ with no consecutive elements and with $\sum_{i\in S}i\equiv 0\pmod 3$. The empty subset contributes one, including at $n = 0$. The objective is a rigorously derived constant-coefficient recurrence with integer coefficients and sufficient initial values, an exact value of a_{12}, and a proof that $a_n/F_{n+2} \to 1/3$, where $F_0 = 0$ and $F_1 = 1$.

The adjacency restriction admits a local include-or-omit decomposition, but the residue increment depends periodically on the index being processed. Weighted generating polynomials and a six-state residue model encode this dependence exactly. The central distinction is between an exact encoding, a finite calculation based on that encoding, and a constant-coefficient recurrence or asymptotic statement valid for every index.

The article will present a partial mathematical result rather than a completed solution. Its main contributions are the explicit rejection of a misleading short recurrence, the reported exact finite value $a_{12} = 128$, and a corrected explanation of the gap-removal weights. The proposed order-twelve recurrence and the limiting density remain separate proof obligations.

Claims discussed: C11, C13

2 Evidence and scope

The finite-count analysis combines saved bit-mask enumeration with an exact-integer dynamic program. Enumeration rejects masks containing adjacent selected positions and sums the one-based selected indices modulo three. The dynamic program records a state (r, b), where $r \in \{0, 1, 2\}$ is the accumulated residue and $b \in \{0, 1\}$ indicates whether the preceding position was selected; its saved calculations extend through $n = 60$.

The symbolic analysis compares two encodings of the same weighted subsets. The weighted recurrence separates subsets according to their largest possible selected element, and root-of-unity averaging extracts residue zero. The alternative encoding removes the mandatory gaps between selected positions, then tracks the positional shift and the normalization of the Gaussian polynomial separately.

The reported computational comparisons concern bounded statements: the initial enumeration through $n = 8$, polynomial agreement through $n = 18$, and recurrence residuals through $n = 48$ using counts through $n = 60$. The synthesis distinguishes these finite outcomes from the proposed matrix characteristic identities and from the unproved asymptotic estimate. No additional computation or independent formal verification is represented as having been performed.

Claims discussed: C10, C12

3 The counting model and residue extraction

The weighted counting polynomial gives an exact reduction of the divisibility condition while retaining the restriction against consecutive choices. For each integer $n \geq 0$, let $\mathcal{A}_n$ be the family of subsets of $\{1, \ldots, n\}$ containing no pair of consecutive integers, and write $s(S) = \sum_{i \in S} i$. The quantity of interest is $a_n=\#\{S\in\mathcal A_n:3\mid s(S)\}$. At $n = 0$, the underlying interval is empty, so $\mathcal A_0=\{\varnothing\}$; with $s(\varnothing)=0$, the empty subset contributes to a_n for every $n \geq 0$.

Define $P_n(q) = \sum_{S \in \mathcal{A}_n} q^{s(S)}$, so that the exponent records the subset sum and the coefficient counts admissible subsets with that sum. The initial polynomials are $P_0(q) = 1$ and $P_1(q) = 1 + q$: the former records only the empty subset, while the latter records the empty subset and $\{1\}$. These initial values are specified separately from the recurrence.

For $n \geq 2$, partition $\mathcal{A}_n$ according to whether a subset contains n. Subsets omitting n are exactly the members of $\mathcal{A}_{n-1}$ and contribute $P_{n-1}(q)$. A subset containing n must omit $n - 1$; deleting n therefore leaves an arbitrary member of $\mathcal{A}_{n-2}$, and conversely adjoining n to any such member

preserves admissibility. Since adjoining n increases the sum by n, this second class contributes $q^n P_{n-2}(q)$, proving $P_n(q) = P_{n-1}(q) + q^n P_{n-2}(q)$ for $n \geq 2$.

At $q = 1$, every admissible subset has weight one, and the weighted recurrence becomes $P_n(1) = P_{n-1}(1) + P_{n-2}(1)$. Its initial values are $P_0(1) = 1$ and $P_1(1) = 2$. With the Fibonacci convention $F_0 = 0$, $F_1 = 1$, and $F_{m+2} = F_{m+1} + F_m$ for $m \geq 0$, these are F_2 and F_3, respectively. Induction therefore gives the exact total $P_n(1) = F_{n+2}$ for every $n \geq 0$.

To extract the required residue class, let ω be a primitive cube root of unity, so $\omega^3 = 1$, $\omega\ne1$, and $1 + \omega + \omega^2 = 0$. For any integer s, the average $(1 + \omega^s + \omega^{2s})/3$ equals one when $3 \mid s$ and zero otherwise. Indeed, divisibility makes all three summands equal to one; in either remaining residue class, the summands are 1, ω, and ω^2 in some order. Applying this identity separately to each subset gives $a_n = \sum_{S \in \mathcal{A}_n} (1 + \omega^{s(S)} + \omega^{2s(S)})/3$.

Interchanging these finite sums yields the root-of-unity filter $a_n = (P_n(1) + P_n(\omega) + P_n(\omega^2))/3 = (F_{n+2} + P_n(\omega) + P_n(\omega^2))/3$. It is the average of the three evaluations, not either nontrivial evaluation alone, that selects sums divisible by three. The counting problem is thus reduced exactly to a known Fibonacci contribution and two explicitly defined weighted evaluations, each governed by the proved polynomial recurrence and its separate initial values.

Claims discussed: C1, C2

4 Exact finite counts and a false short recurrence

Exact finite counting supplies two distinct contributions to the recurrence problem: it rejects a plausible short rule and determines the requested finite value without assuming any general recurrence. The six-state dynamic program retains precisely the information needed to extend an admissible subset: its sum modulo three and whether the most recently processed position was selected. This gives an exact counting procedure, rather than a recurrence inferred from the first few values of a_n.

Let $D_i(r, b)$ count admissible subsets of $\{1, \ldots, i\}$ whose sum has residue $r \in \{0, 1, 2\}$, where $b \in \{0, 1\}$ records whether position i was selected. Initialize $D_0(0, 0) = 1$ and all other states to zero, so that the empty subset is included. On processing position $i \geq 1$, omission is permitted from either previous selection state, giving $D_i(r, 0) = D_{i-1}(r, 0) + D_{i-1}(r, 1)$. Inclusion is permitted only when the preceding position was omitted, giving $D_i(r,1)=D_{i-1}((r-i)\bmod 3,0)$.

These transitions preserve the required interpretation of the states: omission leaves the sum unchanged, whereas inclusion adds the one-based position i and excludes any subset containing $i - 1$. Conversely, every admissible subset has exactly one of these two terminal choices, so the transitions neither omit nor duplicate a subset. The desired count is therefore $a_i = D_i(0, 0) + D_i(0, 1)$. This argument justifies the state-based computation; it does not by itself supply a scalar constant-coefficient recurrence for a_n.

The small cases were also computed by exhaustive bit-mask enumeration, which tests each subset for adjacent selected positions and then tests its one-based sum modulo three. Unlike the dynamic program, this procedure inspects individual subsets rather than aggregating them into six states.

Including the empty subset, the reported exact values are $(a_0, \dots, a_8) = (1, 1, 1, 2, 3, 5, 8, 12, 19)$.

These values decisively reject the tentative rule $a_{n+3} = a_{n+2} + a_n$. Although it agrees with the transitions at $n = 0$ and $n = 1$, at $n = 2$ it would require $a_5 = a_4 + a_2$, whereas the displayed counts give $a_5=5\ne3+1=4$. This is a counterexample to that rule as a recurrence valid from $n = 0$; it says nothing against the possibility of a different constant-coefficient recurrence.

The finite dynamic-programming and weighted-polynomial calculations report $a_{12} = 128$. This value is a result of exact finite computation, not a consequence of the proposed order-twelve recurrence. The reported implementations and calculations have not been independently formally verified, so their agreement should not be assigned the force of a separate proof.

The six-state computation extends through $n = 60$ and reports zero residuals for the order-twelve candidate at every tested starting index $0 \le n \le 48$. That bounded agreement supports retaining the candidate for algebraic investigation, but does not establish it at untested indices. Thus the finite evidence changes the recurrence problem in an asymmetric way: the short rule is eliminated by a single counterexample, while the longer candidate must earn its general validity from a transition identity rather than from its numerical fit.

Claims discussed: C3, C4, C10, C12

5 Gap removal and the corrected Gaussian weight

Gap removal gives an exact encoding of the nonconsecutiveness restriction and identifies the two distinct weight shifts needed for the proposed Gaussian-polynomial representation. The bijection concerns subsets of a fixed cardinality, whereas the Gaussian normalization concerns their positional sums. Keeping these operations separate explains the exponent k^2 and corrects the earlier weight calculation without treating finite polynomial agreement as an all-index proof.

Fix a nonempty admissible subset $1 \le i_1 < \cdots < i_k \le n$, so that $i_{j+1} - i_j \ge 2$, and put $y_j = i_j - (j - 1)$. Then $y_{j+1} - y_j = i_{j+1} - i_j - 1 \ge 1$, while $y_1 \ge 1$ and $y_k \le n - k + 1$. Thus the shifted positions form an ordinary k-subset of $\{1, \dots, n-k+1\}$. Conversely, from any such ordinary subset, the assignment $i_j = y_j + (j - 1)$ gives $i_{j+1} - i_j \ge 2$, $i_1 \ge 1$, and $i_k \le n$; the two assignments are inverse, proving the bijection.

Summing the offsets gives $\sum_{j=1}^{k} i_j = \sum_{j=1}^{k} y_j + k(k-1)/2$. Consequently, gap removal changes each subset's weight by the fixed factor $q^{k(k-1)/2}$, independently of its shifted positions. The allowable cardinalities satisfy $k \le n-k+1$, equivalently $k\leq\lceil n/2\rceil$. The empty subset is handled separately at $k = 0$ and contributes the constant polynomial 1.

For clarity, the partition-normalized Gaussian convention used here is $\genfrac{[}{]}{0pt}{}{m}{k}_q=\sum_{1
y_1<\cdots<y_k\leq m}q^{\sum_{j=1}^k y_j-k(k+1)/2}$ for $0 \le k \le m$. Its exponent removes the baseline sum $1 + \cdots + k = k(k + 1)/2$ of an ordinary subset indexed from 1. Accordingly, restoring the ordinary positional weight requires the factor $q^{k(k+1)/2}$, not $q^{k(k-1)/2}$. The latter factor belongs to gap removal and cannot also serve as the Gaussian normalization.

These two contributions add to $k(k+1)/2+k(k-1)/2 = k^2$, explaining the exponent in the corrected proposed representation $P_n(q)=\sum_{k=0}^{\lceil n/2\rceil}q^{k^2}\genfrac{[}{]}{0pt}{}{n-

k+1}{k}_q$. In this expression the $k = 0$ summand is 1, preserving the empty-subset convention. The correction therefore concerns the explanation of the exponent, not a replacement of the proposed factor q^{k^2}.

Exact integer-coefficient comparisons report agreement between this Gaussian expression and the weighted recurrence for every $0 \le n \le 18$, before any specialization of q. This is corroboration at the polynomial level, but the full all-index Gaussian derivation remains an open obligation in the argument; the finite comparison does not discharge it. The established contribution is the reversible gap-removal encoding and its precise weight shift, which locates the normalization error and gives the alternative representation a concrete combinatorial basis rather than leaving it as a numerical pattern.

Claims discussed: C5, C6, C7

6 The order-twelve candidate and its missing matrix argument

The order-twelve candidate turns the search for a recurrence into a specific question about three-step matrix evolution. The proposed relation is $a_{n+12} - 5a_{n+9} + 2a_{n+6} + 5a_{n+3} + a_n = 0$, with intended range $n \ge 0$. Its largest index offset is twelve, but its nonzero terms occur at intervals of three. It therefore preserves each residue class of indices separately and is distinct from the rejected short rule $a_{n+3} = a_{n+2} + a_n$; the counterexample to that rule does not refute this candidate.

The finite evidence supports precisely this longer relation. Exact six-state dynamic programming, with states recording the sum residue and whether the preceding position was selected, computed counts through $n = 60$ and reported zero residuals for the candidate at every index $0 \le n \le 48$. These are bounded checks of the displayed identity, not a derivation valid for arbitrary indices. Their significance here is that they corroborate the coefficients suggested by the root-of-unity argument while leaving its matrix justification to be supplied.

The proposed derivation separates the ordinary Fibonacci component $P_n(1)$ from the two nontrivial evaluations $P_n(\omega)$ and $P_n(\omega^2)$, where ω is a primitive cube root of unity. It assigns the three-step characteristic polynomial $x^2 - 4x - 1$ to the Fibonacci component and $x^2 - x - 1$ to each nontrivial component. The crucial assertion behind the latter assignment is that every relevant ordered three-step matrix product has trace 1 and determinant -1. These block properties are asserted in the candidate argument, but the products establishing them are not provided.

The algebraic consequence of those assertions can nevertheless be stated conditionally and precisely. Let T denote advancement by three indices, so that $(Tu)_n = u_{n+3}$ for a sequence u. If the Fibonacci component is annihilated by $T^2 - 4T - 1$ and both nontrivial components are annihilated by $T^2 - T - 1$, then their root-of-unity average is annihilated by the product of these two polynomials in T. The polynomial multiplication itself gives $(x^2 - x - 1)(x^2 - 4x - 1) = x^4 - 5x^3 + 2x^2 + 5x + 1$, which yields exactly the proposed recurrence upon replacing x by T. Thus the displayed coefficients follow from the asserted component identities; this algebra does not establish those identities.

The missing bridge is a phase-sensitive matrix calculation. For each starting residue class modulo three, the transition matrices must be multiplied in their actual order of application, and the resulting block matrix must be shown to satisfy the required polynomial identity. Periodicity alone

does not permit reordering these matrices, since they do not generally commute. In particular, if a nontrivial block matrix B were shown to have the asserted trace and determinant, the Cayley-Hamilton identity would give $B^2 - B - I = 0$; applying that identity to the corresponding state evolution would then justify the claimed scalar relation for that phase.

The analogous Fibonacci block identity must also be connected to the same three-index advancement before the annihilating polynomials can be combined.

The intended starting index creates a separate boundary obligation. The weighted recurrence is stated only for $n \geq 2$, with its initial values handled separately, so a matrix argument that reaches backward through an unavailable transition cannot automatically establish the candidate at $n = 0$. A complete derivation must connect the initial states to the ordered blocks and justify every boundary index required by the claimed range. The finite residual check includes $n = 0$, but it does not supply that algebraic connection.

A usable recurrence specification also requires sufficient initial data. Conditional on the proposed relation holding for every $n \geq 0$, the twelve values $a_0, \ldots, a_{11}$ would determine subsequent terms, equivalently providing four starting values for each of the three index classes. The supplied results display values only through a_8, together with a separately reported value at a_{12}, rather than a complete initial list through a_{11}.

The evidence therefore sharpens the recurrence problem from a search for plausible coefficients to a concrete task: establish the ordered block identities, connect them to the initial boundary, and supply the initial data needed to make the resulting relation a complete determination of the sequence.

Claims discussed: C8, C9, C10, C13

7 Asymptotic density and the scope of the result

The root-of-unity representation isolates the precise estimate needed for the proposed density: the Fibonacci contribution is already exact, and only the nontrivial evaluations remain to be controlled. Let ω be a primitive cube root of unity. Using $P_n(1) = F_{n+2}$ in the residue filter gives $\frac{a_n}{F_{n+2}} = \frac{1}{3} + \frac{P_n(\omega)+P_n(\omega^2)}{3F_{n+2}}$. Thus the proposed limiting value is the contribution of the unweighted count, rather than a value inferred from numerical ratios.

Because $P_n(q)$ has integer coefficients and $\omega^2 = \overline{\omega}$, the two nontrivial evaluations satisfy $P_n(\omega^2) = \overline{P_n(\omega)}$. Their sum is therefore $2\,\mathrm{Re}\,P_n(\omega)$, but this reality condition supplies no bound on its magnitude. The remaining asymptotic obligation is to prove $P_n(\omega) + P_n(\omega^2) = o(F_{n+2})$ as $n \to \infty$; by the displayed identity, this is equivalent to $a_n/F_{n+2} \to 1/3$. A bound $|P_n(\omega)| = o(F_{n+2})$ would suffice, since conjugacy gives $|P_n(\omega) + P_n(\omega^2)| \leq 2|P_n(\omega)|$.

This growth estimate is logically distinct from establishing the proposed order-twelve recurrence. Even if that recurrence were proved globally, its existence alone would not identify which characteristic contributions occur in this particular sequence, their coefficients, or their size relative to the Fibonacci denominator. A recurrence-based asymptotic argument would therefore need control of the relevant roots and coefficients, while a transfer-matrix argument would need a justified bound

for the nontrivial modes. The available argument contains no completed estimate of either kind; finite ratios cannot replace such control.

The bounded computations have a narrower role: they report $a_{12} = 128$ and support the order-twelve candidate on the tested indices, without establishing its global validity or the density limit. Agreement among these calculations and mathematical accounts is corroboration, not an independent proof of the missing assertions. Nor is the recurrence specification complete: a sufficient displayed initial-value list and algebraically justified treatment of the initial boundary remain outstanding.

The resulting contribution is therefore a reduction of the problem through exact counting encodings, together with a reported exact finite count and a concrete recurrence candidate; for the density question, the reduction identifies the nontrivial root-of-unity contribution, rather than the Fibonacci main term, as the decisive object of proof.

Claims discussed: C2, C4, C8, C11, C12, C13

8 Verification status and limitations

- The reported exact computations are bounded results and have not been independently rerun or formally verified in this synthesis; their reported scope is recorded in C3, C4, C7, C10, and C12.
- The Gaussian-polynomial representation has a supported bijective explanation and a corrected exponent convention, but the supplied assessments conflict about whether its universal derivation is complete. C6 preserves that uncertainty instead of using finite agreement to settle it.
- The global recurrence and limiting density are not established by the available evidence. The specific missing matrix, boundary, initial-value, and asymptotic steps are localized in C9, C11, and C13.
- The evidence index is a bounded selection, not an exhaustive review of all saved material. No assertion of complete semantic coverage or of access to omitted full notes is made.

9 Open obligations

- Complete the all-index Gaussian-polynomial derivation under the explicitly stated subset-indexing convention, including the empty-subset and size-bound cases; resolve the conflicting proof assessments attached to C6.
- Display the exact ordered three-step transfer products for every phase and verify the claimed characteristic identities, rather than assuming that periodic transition matrices commute.
- Derive the proposed integer-coefficient recurrence for every intended starting index from the justified matrix identities and the root-of-unity filter.

- Handle the initial boundary algebraically and supply a complete sufficient initial-value list, without filling missing entries by assuming the conjectural recurrence.
- Prove a bound on the nontrivial root-of-unity evaluations that makes their combined contribution $o(F_{n+2})$, and then deduce the limiting ratio.

10 Conclusion

The local combinatorial decomposition gives an exact weighted recurrence and a residue filter. Gap removal supplies a second exact combinatorial encoding, with the corrected normalization explaining the proposed Gaussian weight.

The finite evidence reports $a_{12} = 128$ and decisively rejects $a_{n+3} = a_{n+2} + a_n$. It also supports, but does not prove, the order-twelve every-third-term candidate.

A complete solution still requires the phase-ordered matrix derivation, sufficient initial data with boundary justification, and a separate asymptotic estimate. The article should therefore conclude with a precisely bounded partial result, not a claimed proof of the requested recurrence or density.

Claims discussed: C2, C3, C4, C5, C6, C8, C9, C10, C11, C13

Claim status and evidence

10.1 C12 – Limitation

The available mathematical reports and finite computations are not independently formally verified; agreement between reports does not establish the missing global proofs. The supplied reports explicitly identify deterministic fixture narratives, finite exact computations, and incomplete verification. The article may explain supported elementary arguments and report bounded numerical outcomes, but it must not convert agreement or a favorable review into a proof of an all-index statement. *Evidence:* session:1aabbfc0-e642-4581-a96f-2e33ac17ee19, session:6c552b3c-44e4-4b57-8b93-e5a645d057d1, session:7c70ca6e-3fb2-4cda-b75a-72702ee10d97

10.2 C13 – Unresolved

A sufficient initial-value list for the proposed order-twelve recurrence, together with algebraically justified boundary handling, remains an uncompleted requirement. The indexed material gives a_0 through a_8 and reports a_{12}, but it does not supply a complete displayed list of a_0 through a_{11}. The saved obligations also require exact treatment of the initial boundary rather than reliance only on dynamic-programming checks. Missing values will not be inferred from an unproved recurrence. *Evidence:* artifact:6c552b3c-44e4-4b57-8b93-e5a645d057d1:5e15b337-adf1-483c-8ba6-7cd2de5f500b, artifact:7c70ca6e-3fb2-4cda-b75a-72702ee10d97:516c2288-3458-40b2-9e02-87a918c35a3a

10.3 C10 – Unresolved

A saved exact six-state dynamic program computes counts through $n = 60$ and reports vanishing residuals for the order-twelve candidate through $n = 48$. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. The state records the current sum residue and whether the last position was selected. Omitting the next position resets the selection bit; including it is allowed only after an omission and adds that position's residue. This supports the reported finite check, not a recurrence for arbitrary n; the status here concerns the bounded reported computation, subject to C12. *Evidence:* artifact:6c552b3c-44e4-4b57-8b93-e5a645d057d1:5e15b337-adf1-483c-8ba6-7cd2de5f500b, session:6c552b3c-44e4-4b57-8b93-e5a645d057d1, strategy:5c8ce76f-4965-434b-8862-841c92de89e4

10.4 C11 – Unresolved

The requested limit $a_n/F_{n+2} \to 1/3$ is unresolved in the available argument; the root-of-unity approach still requires $P_n(\omega) + P_n(\omega^2) = o(F_{n+2})$. Dividing the residue filter by F_{n+2} isolates the desired main term and the two nontrivial evaluations. Those evaluations are complex conjugates because the polynomial has integer coefficients, but conjugacy alone does not bound their size. The saved evidence contains no completed estimate proving that their contribution is negligible. *Evidence:* artifact:1aabbfc0-e642-4581-a96f-2e33ac17ee19:ccc1045f-479d-4018-b8ed-42a60b94e567, artifact:6c552b3c-44e4-4b57-8b93-e5a645d057d1:5e15b337-adf1-483c-8ba6-7cd2de5f500b, artifact:7c70ca6e-3fb2-4cda-b75a-72702ee10d97:516c2288-3458-40b2-9e02-87a918c35a3a

10.5 C2 – Unresolved

Writing $P_n(q) = \sum_S q^{\sum_{i \in S} i}$ over admissible subsets gives $P_0(q) = 1$, $P_1(q) = 1 + q$, and $P_n(q) = P_{n-1}(q) + q^n P_{n-2}(q)$ for $n \geq 2$. Also, $P_n(1) = F_{n+2}$ and $a_n = (P_n(1) + P_n(\omega) + P_n(\omega^2))/3$ for a primitive cube root ω. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. An admissible subset either omits n, leaving a subset of the preceding interval, or includes n, forcing omission of $n - 1$ and contributing weight q^n. At $q = 1$, this recurrence and the initial values give the Fibonacci count. For an integer exponent s, the average $(1 + \omega^s + \omega^{2s})/3$ is one when 3 divides s and zero otherwise, so averaging the evaluations counts precisely the required residue. *Evidence:* strategy:00469f7f-ba53-4a12-944d-2d3295e48d1f, artifact:1aabbfc0-e642-4581-a96f-2e33ac17ee19:ccc1045f-479d-4018-b8ed-42a60b94e567

10.6 C1 – Unresolved

The problem concerns one-based subsets of $\{1, \ldots, n\}$ with no consecutive elements, includes the empty subset, and requests a recurrence, sufficient initial values, a_{12}, and a Fibonacci-normalized limiting density. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. These are the stated

counting conventions and objectives, not conclusions about whether the requested proof has been completed. The convention at $n = 0$ makes the empty subset the unique admissible subset. *Evidence:* project:57d67c57-15dc-433d-828a-7953f7986ff8, artifact:1a9a0c44-48ae-49e7-8bea-0e8d47297f9d:1cc6ec50-87c0-46f4-b1dd-a82857d74f9e

10.7 C3 – Unresolved

The reported exact values $(a_0, \ldots, a_8) = (1, 1, 1, 2, 3, 5, 8, 12, 19)$ refute $a_{n+3} = a_{n+2} + a_n$: at $n = 2$, $a_5 = 5$ but $a_4 + a_2 = 4$. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. The saved enumeration includes the empty subset, tests adjacency, and uses one-based positions for the sum. The proposed rule agrees with its first two tested transitions but fails on the displayed third transition. A single exact counterexample disproves that rule without implying anything against the existence of a different recurrence. *Evidence:* artifact:1a9a0c44-48ae-49e7-8bea-0e8d47297f9d:1cc6ec50-87c0-46f4-b1dd-a82857d74f9e, session:1a9a0c44-48ae-49e7-8bea-0e8d47297f9d

10.8 C4 – Unresolved

The saved finite exact calculations report $a_{12} = 128$. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. The value is reported by the finite dynamic-programming and weighted-polynomial accounts. It is a bounded exact-computation result, not a consequence of a proved all-index recurrence. Its evidentiary scope remains subject to the verification limitation in C12. *Evidence:* artifact:1aabbfc0-e642-4581-a96f-2e33ac17ee19:ccc1045f-479d-4018-b8ed-42a60b94e567, artifact:f7301d22-fd90-4675-95cd-17b006262ef9:02ca743b-d66f-41c0-8692-34579ea5adfa

10.9 C5 – Unresolved

For a nonconsecutive subset $1 \le i_1 < \cdots < i_k \le n$, the substitution $y_j = i_j - (j - 1)$ is a bijection to ordinary k-subsets of $\{1, \ldots, n - k + 1\}$ and satisfies $\sum_j i_j = \sum_j y_j + k(k - 1)/2$. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. The original gaps are at least two, so subtracting successive offsets leaves strictly increasing positions within the stated bounds. Conversely, $i_j = y_j + (j - 1)$ restores gaps of at least two and keeps the last position at most n. Summing the offsets gives the stated weight shift; these are the concrete forward and inverse maps supplied by the evidence. *Evidence:* strategy:9ad51640-ff9a-4d22-a928-518f032003bf, artifact:f7301d22-fd90-4675-95cd-17b006262ef9:02ca743b-d66f-41c0-8692-34579ea5adfa, artifact:7c70ca6e-3fb2-4cda-b75a-72702ee10d97:516c2288-3458-40b2-9e02-87a918c35a3a

10.10 C6 – Unresolved

The corrected proposed Gaussian-polynomial representation is $P_n(q)=\sum_{k=0}^{\lceil n/2\rceil}q^{k^2}\genfrac{[}{]}{0pt}{}{n-k+1}{k}_q$. Its explanation requires the standard

ordinary-subset offset $k(k+1)/2$, not $k(k-1)/2$; the indexed reviews nevertheless retain a complete all-index derivation as an open obligation. Under the supplied partition-normalized convention, \$\genfrac{[}{]}{0pt}{}{m}{k}_q\$ enumerates ordinary subsets with exponent $\sum_j y_j - k(k+1)/2$. Restoring that baseline and then the gap-removal shift gives $k(k+1)/2 + k(k-1)/2 = k^2$. This repairs the earlier explanatory error, but the records disagree on whether the symbolic identity is fully established; the plan therefore preserves the stated all-index proof obligation rather than presenting review agreement as verification. *Evidence:* artifact:f7301d22-fd90-4675-95cd-17b006262ef9:02ca743b-d66f-41c0-8692-34579ea5adfa, artifact:7c70ca6e-3fb2-4cda-b75a-72702ee10d97:516c2288-3458-40b2-9e02-87a918c35a3a, session:7c70ca6e-3fb2-4cda-b75a-72702ee10d97

10.11 C7 – Unresolved

Saved exact integer-polynomial comparisons report agreement between the Gaussian-polynomial expression and the weighted recurrence for $0 \le n \le 18$. Material evidence was omitted from the extracted index; this claim cannot be treated as established until the complete frozen evidence is reviewed. The comparisons are described as exact coefficient calculations before specialization of q, with the $k = 0$ contribution handled as one. They corroborate the corrected expression over the stated finite range. They do not independently prove the identity for all indices or the density limit. *Evidence:* artifact:f7301d22-fd90-4675-95cd-17b006262ef9:02ca743b-d66f-41c0-8692-34579ea5adfa, artifact:7c70ca6e-3fb2-4cda-b75a-72702ee10d97:516c2288-3458-40b2-9e02-87a918c35a3a

10.12 C8 – Conjecture

The proposed recurrence is $a_{n+12} - 5a_{n+9} + 2a_{n+6} + 5a_{n+3} + a_n = 0$ for $n \ge 0$; its global validity remains conjectural in the supplied evidence. The candidate is explicitly stated and is consistent with the reported finite residual calculations. Its largest index offset is twelve, while its nonzero terms are separated by three. The saved material does not complete the phase-ordered matrix argument needed to establish the assertion for every n. *Evidence:* artifact:1aabbfc0-e642-4581-a96f-2e33ac17ee19:ccc1045f-479d-4018-b8ed-42a60b94e567, artifact:6c552b3c-44e4-4b57-8b93-e5a645d057d1:5e15b337-adf1-483c-8ba6-7cd2de5f500b

10.13 C9 – Unresolved

The proposed matrix derivation assigns block characteristic polynomial $x^2 - x - 1$ to the nontrivial cube-root modes and $x^2 - 4x - 1$ to the Fibonacci component. The ordered products, their asserted trace and determinant, and the initial boundary remain unproved in the saved argument. The saved candidate argument asserts trace one and determinant minus one for the nontrivial three-step products, then combines their characteristic polynomial with the Fibonacci block polynomial. Its reviews require explicit products for each phase because the transition matrices do not generally commute. They also identify an initial-boundary issue because the weighted recurrence was stated only for $n \ge 2$. *Evidence:* artifact:1aabbfc0-e642-4581-a96f-2e33ac17ee19:ccc1045f-479d-4018-

b8ed-42a60b94e567, artifact:6c552b3c-44e4-4b57-8b93-e5a645d057d1:5e15b337-adf1-483c-8ba6-7cd2de5f500b

Authorship note

AI-assisted synthesis from saved research; not independently peer reviewed.